GradPilot AI – Multi-Agent Autonomous Workflow Architecture

1. Overview

GradPilot AI is a multi-agent system designed to automate academic workflows for graduation students. It leverages Retrieval-Augmented Generation (RAG) to deliver accurate, syllabus-aligned responses while executing multiple steps autonomously — from query understanding to study planning.

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2. System Architecture

User Query
↓
Query Agent
↓
Retrieval Agent (FAISS Vector DB)
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Context Builder
↓
Content Agent (LLM)
↓
Planner Agent
↓
Validator Agent
↓
Final Output (Notes + Timetable + Audit Trail)

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3. Agent Roles

Query Agent
Processes and refines the user input to understand intent.

Retrieval Agent
Fetches relevant academic content from a vector database built using syllabus-based resources.

Content Agent
Generates structured explanations and exam-focused short notes using LLM.

Planner Agent
Creates personalized study schedules based on user queries.

Validator Agent
Checks response quality and ensures completeness. If output is insufficient, regeneration is triggered.

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4. RAG Implementation

The system uses Retrieval-Augmented Generation to reduce hallucinations. Instead of generating responses purely from the model, it retrieves relevant academic content and feeds it into the LLM for context-aware and accurate outputs.

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5. Tool Integration

- FAISS → Vector database for semantic search
- OpenAI API → Language model for generation
- Streamlit → User interface

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6. Error Handling & Auditability

- Outputs are validated for completeness
- Regeneration triggered if quality is low
- Full workflow steps are displayed as an audit trail

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7. Key Innovation

- Multi-agent coordination for academic workflows
- End-to-end automation from query to planning
- Reduced hallucination using RAG